B1. Field Experiment Test Plan

October 10, 2026 — Prepared by Attractor Co., Ltd.

> This document is the Milestone 1 deliverable "B1. Field Experiment Test Plan" and satisfies acceptance criterion B2 (it defines at least six specific objectives, test scenarios, and expected results).

Purpose and Scope

This document corresponds to the Milestone 1 ("System Study and Test Content for the Field Experiment") acceptance criterion "Specifically define the test items for the field experiment." It sets out the framework of test items, methods, and pass criteria for the field tests to be conducted in M3 ("Conducting the Field Experiment") and for the preliminary verification to be conducted beforehand in a closed environment in M2 ("System Preparation for the Field Experiment").

The scope covers the following three areas.

- Transmission: transmission of block information (block headers, etc.) from the broadcaster via IPDC
- Reception: reception and synchronization of block information by multiple receivers
- Authentication: verification of certificate authenticity in an offline environment

This document defines the test items. Details of the schedule, organization, and equipment will be specified separately in M2 ("System Preparation for the Field Experiment").

Test Design Policy

The tests consist of the minimum set of items needed to achieve the objectives. Only what cannot be confirmed without broadcast waves is tested in the field; what can be confirmed without broadcast waves, such as software correctness and receiver processing capacity, is verified in advance in the M2 closed environment.

The field test consists of a single continuous transmission received simultaneously at three sites, and Tests 1 to 4 are all judged from the logs obtained from this transmission. Repeated runs with different radio conditions, bundle sizes, or terminal models are not performed in the field, and a single model of verification terminal is used.

Test 7 determines the bundle size and overlap width used in the field test and must therefore be completed before the field test.

List of Test Items

No.	Phase	Category	Test Item	Summary
Test 1	M3 Field	Transmission	IPDC transmission check	Can block header information be transmitted correctly via IPDC from the broadcaster's facilities?
Test 2	M3 Field	Reception	Simultaneous reception at three sites	Can block information be received correctly and simultaneously at three sites with different reception

No.	Phase	Category	Test Item	Summary
				conditions?
Test 3	M3 Field	Reception	Gap filling through overlapping transmission	When one bundle is missing, can it be filled without gaps from the adjacent bundles?
Test 4	M3 Field	Authentication	Offline certificate verification and time required	Can certificate authenticity be judged without an Internet connection, within a practical time?
Test 5	M2 Closed environment	Transmission	Correctness of transmitted data generation	Is transmitted data generated correctly even for blocks with many transactions and when a rollback occurs?
Test 6	M2 Closed environment	Authentication	Handling of invalid and missing data	Are fake chains and invalid signatures rejected, and are gaps distinguished from tampering?
Test 7	M2 Closed environment	Reception	Determining bundle size and overlap width	Can a bundle size and overlap width suited to the receiver's recovery processing capacity be determined?

Details of Each Test Item (Objectives, Scenarios, and Expected Results)

No.	Objective	Test Scenario	Expected Result	Related KPI
Test 1	Confirm that block information can be transmitted correctly via IPDC from broadcast	Transmit signed block headers in bundles from the broadcaster's transmission facilities in one continuous run. Compare the transmission log with the	Every transmitted block header matches the content of the reception logs	Block header transmission success rate

No.	Objective	Test Scenario	Expected Result	Related KPI
	facilities	reception logs at each site in Test 2		
Test 2	Confirm that simultaneous distribution to multiple sites works and how differences in reception conditions affect reception	Receive the Test 1 transmission simultaneously with receivers installed at three sites with different reception conditions (e.g., good reception / weak-signal area far from the transmitter / indoors), and record the number of blocks received at each site	All blocks are received at the site with good reception. At the weak-signal and indoor sites, the reception success rate exceeds a threshold agreed in advance (to be confirmed)	Block header transmission success rate
Test 3	Confirm under real broadcast conditions that redundant transmission with the overlapping sliding window compensates for bundles that are missed	Partway through the Test 1 transmission, intentionally skip transmitting exactly one bundle via the sender's settings. Compare the blocks held by the receiver at each site with the list of expected heights in the manifest	All blocks contained in the skipped bundle are filled from the adjacent bundles	Block header transmission success rate
Test 4	Confirm that certificate authenticity verification completes in a practical time without an Internet connection	Prepare a legitimate certificate issued for the experiment and a certificate with tampered content. On the network-isolated verification terminal at each site, match each against the received block	The legitimate certificate is judged "authentic" and the tampered certificate "rejected." The average verification time is within the acceptable range (e.g., a few seconds; to be confirmed)	Verification latency, number of certificate issuance transactions, average transaction cost

No.	Objective	Test Scenario	Expected Result	Related KPI
		information and measure the time from certificate presentation to judgment. When issuing the certificates, record the number of issuance transactions and the ADA cost		
Test 5	Confirm that transmitted data is generated correctly from all transactions and that already-transmitted data is not rewritten on rollback	For blocks with 0, 1, an even number, an odd number, and more than 100 transactions, recompute block_body_hash with an independent implementation and compare. In addition, simulate a rollback and compare the header files and manifests before and after reconstruction	block_body_hash matches in all cases (for more than 100 transactions, it matches the value computed from all transactions, not just the first 100). After a rollback, already-transmitted headers are not overwritten and are recorded as a separate series	—
Test 6	Confirm that fake data that self-consistency checks alone cannot detect is rejected, and that data that was not received is distinguished from tampering	Feed the receiver the following five conditions: a self-consistent fake block header sequence created from arbitrary initial values, data without a signature, data signed with a different key, data with one byte of the signature altered, and data with two consecutive bundles missing	All four fake-data conditions are rejected, with the reason (invalid signature / anchor mismatch / chain inconsistency) recorded separately. Two consecutive missing bundles are detected as a gap and reported as "not received," not as tampering	—
Test 7	Determine the	Feed TS generated by our	The relationship	—

No.	Objective	Test Scenario	Expected Result	Related KPI
	bundle size and overlap width suited to the receiver's processing capacity and use them as the transmission settings for the field test	IPDC system directly into the receiver used in the experiment, without going through broadcast waves, and measure object recovery processing time and miss rate while varying the bundle size (e.g., 1, 10, 30, and 60 blocks)	between bundle size and processing time is understood, and a bundle size and overlap width with sufficient margin relative to the transmission cycle are determined	

Correspondence to Acceptance Criteria (B2)

The tables above define the objectives, test scenarios, and expected results for seven items, Tests 1 to 7, which satisfies acceptance criterion B2: "Define at least six specific objectives, test scenarios, and expected results." Of these, Tests 5 to 7 are based on issues identified by source-reviewing the previous project's implementation and on the receiver-side recovery processing constraints observed in an earlier experiment. Of the proposal's KPIs, the block header transmission success rate is measured in Tests 1 to 3, and the verification latency, number of certificate issuance transactions, and average transaction cost are measured in Test 4.

Pass or fail for each test is judged by whether the result matches the "Expected Result" in the table above. Numerical thresholds (reception success rate, acceptable latency, and other items marked "to be confirmed" in this document) will be finalized once the broadcaster and equipment are confirmed, and this document will be revised accordingly.

The results of the closed-environment preliminary verification (Tests 5 to 7) will be published in M2, and the results of the field tests (Tests 1 to 4) after M3 is conducted, each on GitHub as Evidence of Completion.

Summary

This document defined the tests for the field experiment as a minimum set of seven items needed to achieve the objectives. Tests 1 to 4 are field tests that use broadcast waves and are conducted as a single continuous transmission received simultaneously at three sites. Transmission, reception, gap filling, and offline certificate verification are all judged from the logs obtained from that transmission, so no repeated runs under different conditions are performed in the field.

Tests 5 to 7 are preliminary verification conducted without broadcast waves in the M2 closed environment. They confirm that transmitted data is generated correctly and that fake data and invalid signatures are rejected while gaps are distinguished from tampering, and they determine a bundle size and overlap width suited to the receiver's processing capacity. The values determined in Test 7 are used as the transmission settings for the field test.

All four KPIs in the proposal can be measured in the field tests. The block header transmission success rate is measured in Tests 1 to 3, and the verification latency, number of certificate issuance transactions, and average transaction cost are measured in Test 4.

Numerical pass thresholds, the installation locations and reception conditions of the three sites, the broadcaster and channel to be used, the

timing of the tests, and the type of certificate will be set once the broadcaster and equipment are confirmed, and this document will be revised accordingly.